

Netzwerke und Schaltungen 1

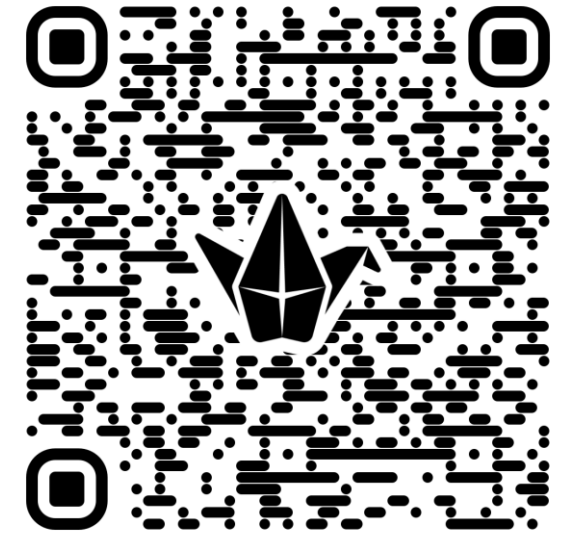
Übung 01

Vektoren und Integralrechnung

<https://eputzi.github.io/courses/nus1/index.html>



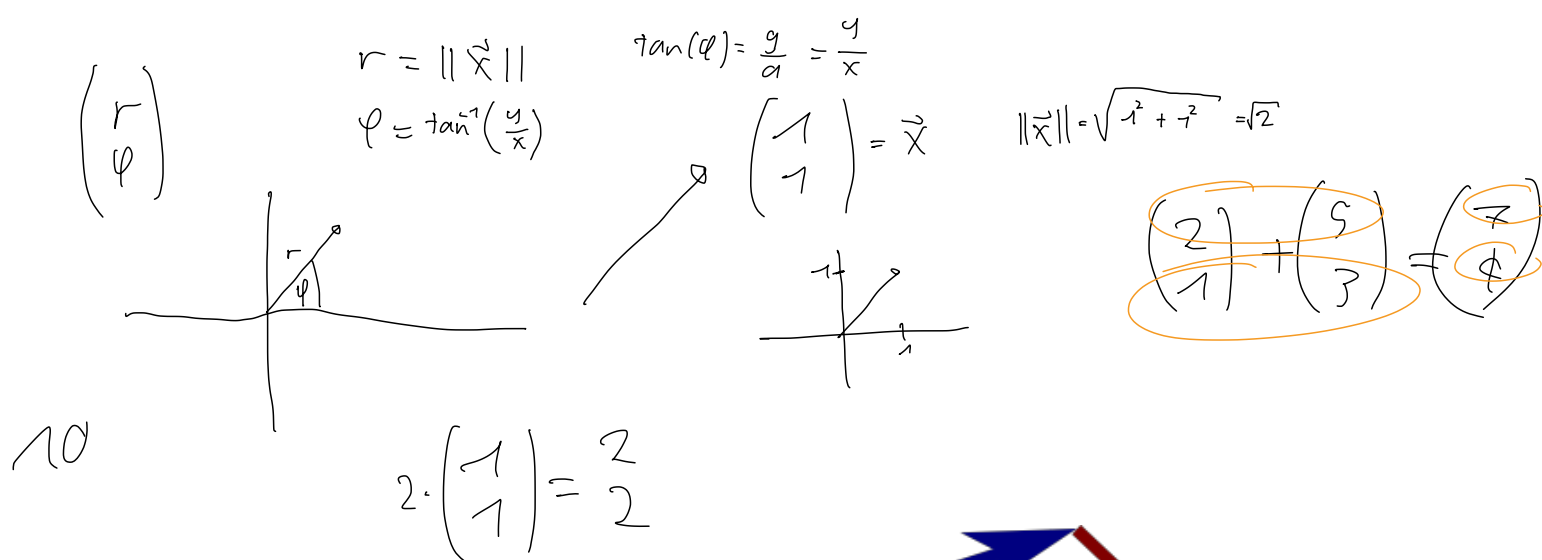
Webseite



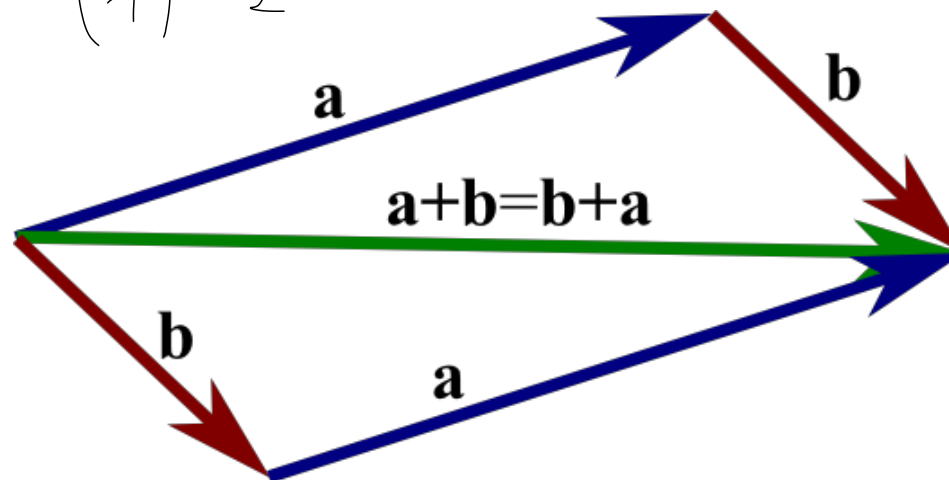
Fragen

NUS1

Vektoren



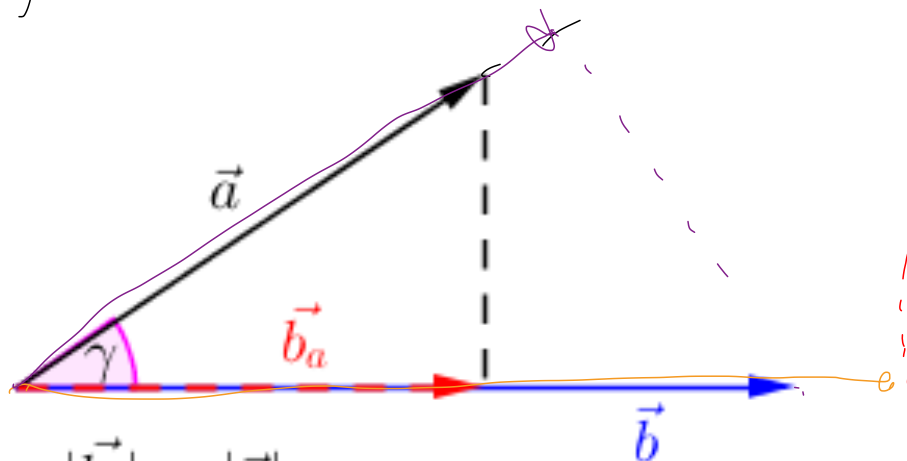
- Skalar (einfach eine Zahl)
- Vektor (Betrag und Richtung)
- Operationen
 - Addition
 - Streckung/Stauchung
 - Skalarprodukt
 - Kreuzprodukt



Skalarprodukt/Kreuzprodukt

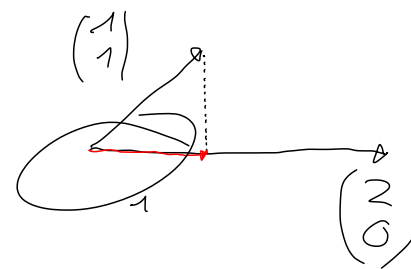
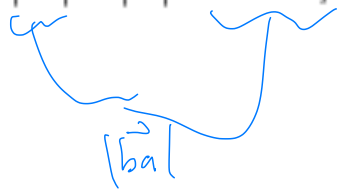
$$\left\| \begin{pmatrix} 4 \\ 3 \end{pmatrix} \right\| = 5$$

$$\frac{1}{5} \cdot \begin{pmatrix} 4 \\ 3 \end{pmatrix}$$



$$|\vec{b}_a| = |\vec{a}| \cdot \cos \gamma$$

$$\vec{a} \cdot \vec{b} = |\vec{a}| \cdot |\vec{b}| \cdot \cos \gamma = |\vec{a}| \cdot \cos \gamma \cdot |\vec{b}| = |\vec{b}_a| \cdot |\vec{b}|$$



$$1 \cdot 2 + 1 \cdot \underbrace{0}_0 = 2$$

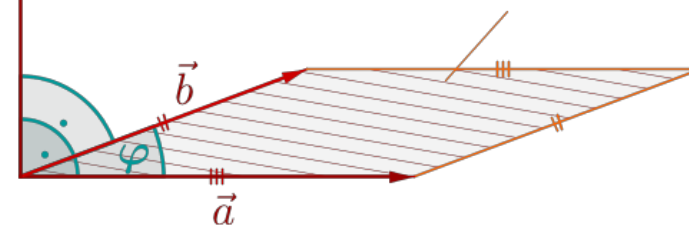
$$\frac{2}{2} = 1$$

$$\begin{pmatrix} 1 \\ 0 \\ 1 \\ 1 \\ 1 \\ 0 \end{pmatrix} \times \begin{pmatrix} 2 \\ 1 \\ 1 \\ 1 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \cdot 1 - 1 \cdot 1 \\ 1 \cdot 2 - 1 \cdot 1 \\ 1 \cdot 1 - 0 \cdot 2 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}$$

$$\vec{c} = \vec{a} \times \vec{b} \neq \vec{b} \times \vec{a}$$

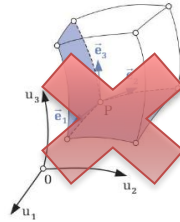
$$\begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = 0$$

$$A = |\vec{c}| = |\vec{a}| \cdot |\vec{b}| \cdot \sin(\varphi)$$



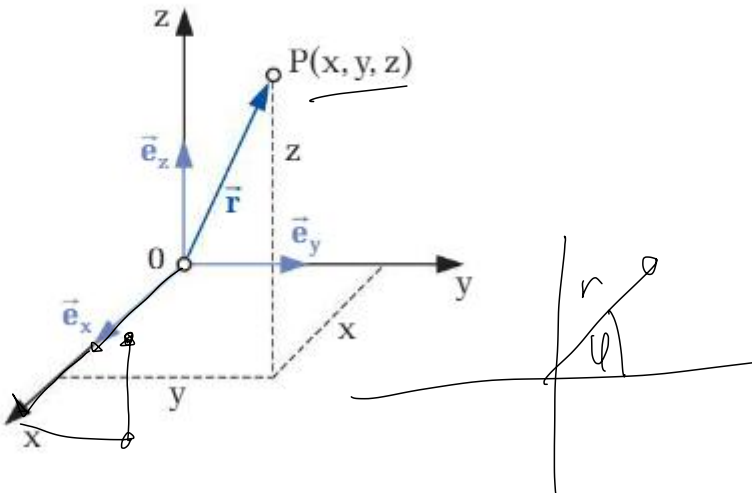
Koordinatensysteme

Forget krummlinige Koordinatensysteme

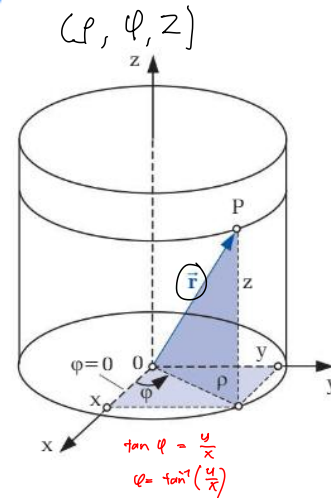


Das kartesische Koordinatensystem

$$\begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$$

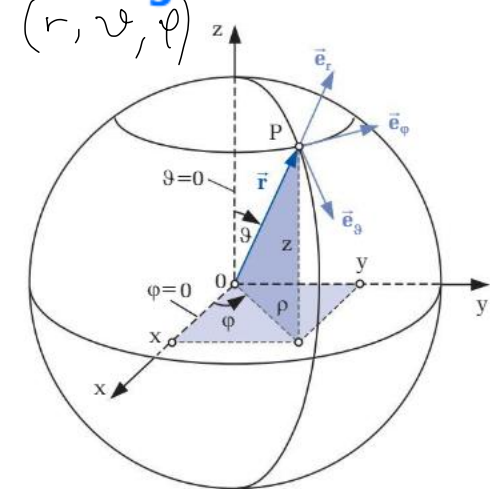


Die Zylinderkoordinaten



$$\begin{aligned} x &= \rho \cos \varphi & 0 \leq \rho < \infty \\ y &= \rho \sin \varphi & \text{mit } 0 \leq \varphi < 2\pi \\ z &= z. \end{aligned}$$

Die Kugelkoordinaten



$$\begin{aligned} x &= r \sin \vartheta \cos \varphi & 0 \leq r < \infty \\ y &= r \sin \vartheta \sin \varphi & \text{mit } 0 \leq \vartheta \leq \pi \\ z &= r \cos \vartheta & 0 \leq \varphi < 2\pi \end{aligned}$$

Kartesisch -> Zylinder

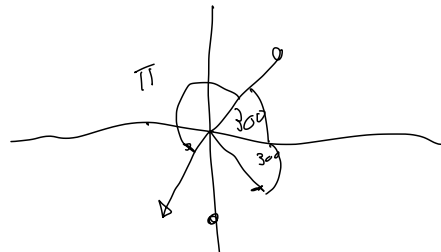
$$P = (3, 3 * \sqrt{3}, 5)$$

$$r = \sqrt{3^2 + (3\sqrt{3})^2} = \sqrt{9 + 27} = 6$$

$$\varphi = \tan^{-1}\left(\frac{3\sqrt{3}}{3}\right) = \tan^{-1}(\sqrt{3}) = 60^\circ$$

$$P = (6, 60^\circ, 5)$$

$$\tan^{-1}\left(\frac{-\sqrt{3}}{-1}\right) = 30^\circ$$



Kartesisch -> Kugel

$$P = (1, \sqrt{3}, 2 * \sqrt{3})$$

$$r = \sqrt{1^2 + \sqrt{3}^2 + (2\sqrt{3})^2}$$

$$= \sqrt{1 + 3 + 12} = 4$$

$$\cos(\varphi) = \frac{z}{r} \rightarrow \varphi = \cos^{-1}\left(\frac{z}{r}\right) = 30^\circ$$

$$\cos^{-1}\left(\frac{2\sqrt{3}}{4}\right) = 30^\circ$$

$$\varphi = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{\sqrt{3}}{1}\right) = 60^\circ$$

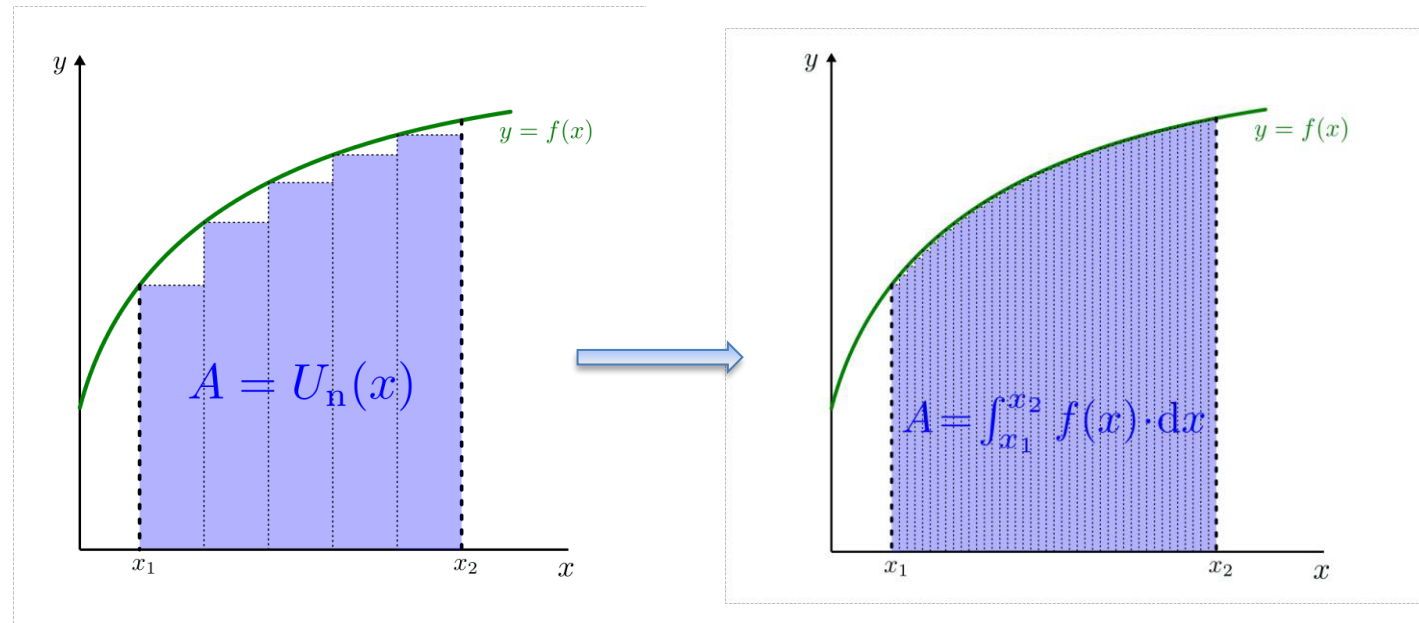
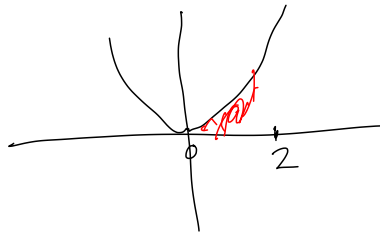
$$P(4, 30^\circ, 60^\circ)$$

Integral

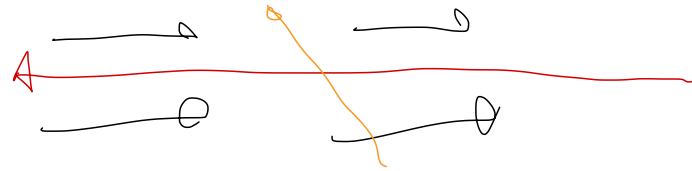
- Integral wie eine Summe denken

$$f(x) = x^2$$

$$\int_0^2 x^2 dx = \left[\frac{x^3}{3} \right]_0^2 = \frac{2^3}{3} - \frac{0^3}{3} = \frac{8}{3}$$



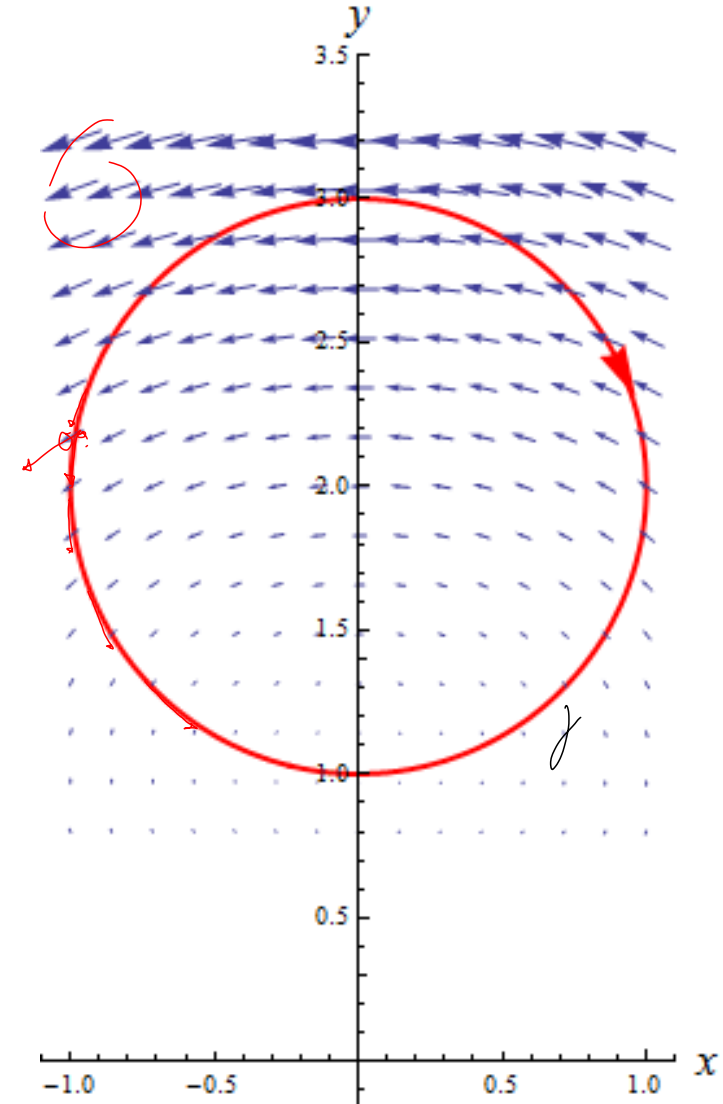
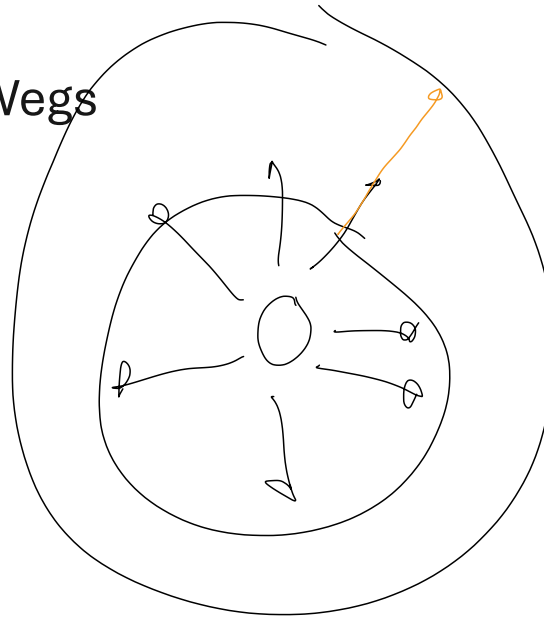
Linienintegral



- Man summiert entlang des Wegs

$$\int_{\gamma} \overrightarrow{F(x, y)} \cdot d\vec{s}$$

$$\int_0^{2\pi} \underbrace{f(\gamma(x))}_{\text{red circle}} \underbrace{\gamma'(x)}_{\text{red underline}} dx$$



Flächen und Volumenintegrale

- Keine Angst, ihr müsst keine komplett ausrechnen (in NUS1)
- Immer Möglich zu vereinfachen

$$Q = \iint_A \sigma dA. \quad Q = \iiint_V \rho dV.$$

#04 – Vereinfachung Wegintegral

(<https://www.youtube.com/watch?v=A76JpF7ktCl>)

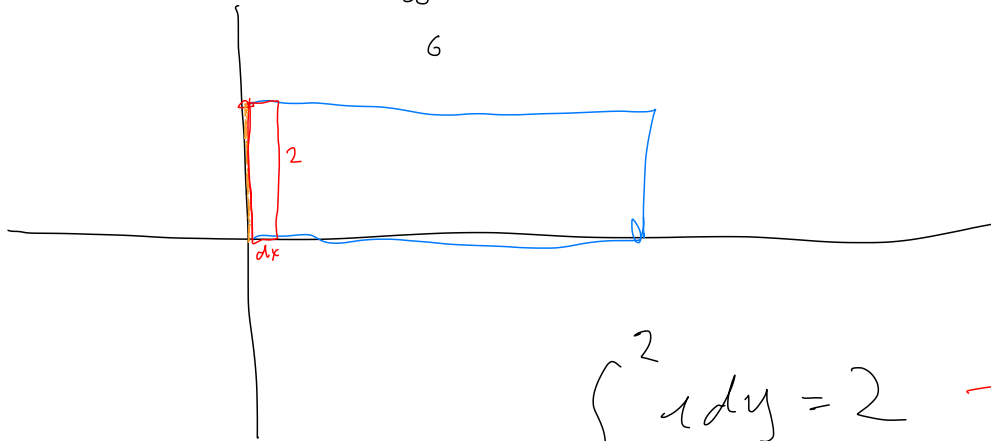
#03 – Vereinfachung Oberflächenintegral

(<https://www.youtube.com/watch?v=NIgysPeUliY>)

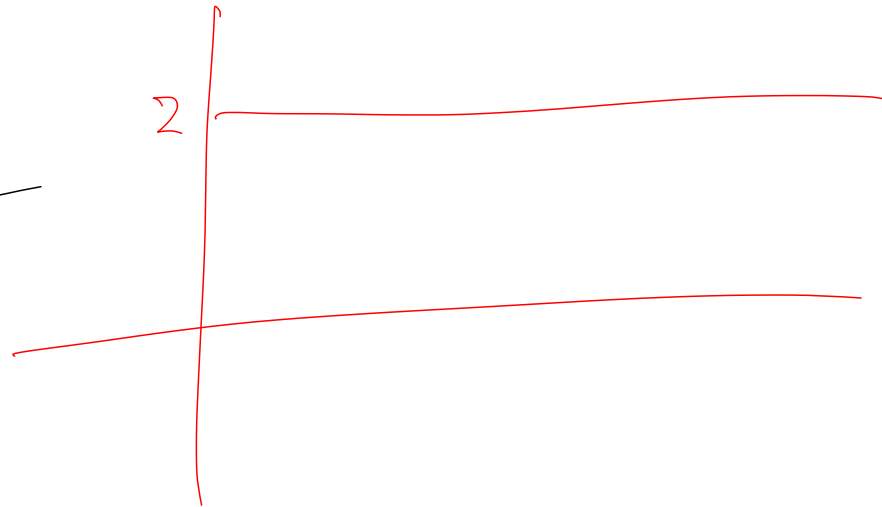
Rechteckfläche berechnen

$$A = \int_0^3 \int_0^2 1 \, dy \, dx$$

$\int_0^3 2 \, dx$ $\int_0^2 1 \, dy = 2$ $\int_0^3 2 \, dx$



$$\int_0^2 1 \, dy = 2$$



$$f(x) = 2$$

Takeaways

Drei Dinge, die heute sitzen sollten. (Mehr oder weniger)

01

Vektorrechnung

02

Koordinatensysteme kennen

03

**Evtl. verstehen was Flächen-
und Kurvenintegrale sind**

Fragen?